

$$\int_{-\infty}^{\infty} \delta(\omega - k) f(\omega) d\omega = f(k)$$

$$\frac{1}{j} = \frac{j}{j^2} = -j$$

QUIZ I - FOURIER TRANSFORM

1. What is the Fourier transform of the following CT signal?

$$x(t) = \sin(t) + 0.5 \sin(20t)$$

$$= \frac{1}{2j} (e^{jt} - e^{-jt}) + \frac{0.5}{2j} (e^{j20t} - e^{-j20t})$$

(A) $\pi(\delta(\omega - 1) + \delta(\omega + 1) + 0.5\delta(\omega - 20) + \delta(\omega + 20))$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{X}(j\omega) e^{-j\omega t} d\omega$$

(B) $\pi j(\delta(\omega + 1) - \delta(\omega - 1) + 0.5(\delta(\omega + 20) - \delta(\omega - 20)))$

(C) $\delta(\omega - 1) + 0.5\delta(\omega - 20)$

$$x(t) = \frac{1}{2\pi} \int_{-\infty}^{\infty} \hat{X}(j\omega) \overbrace{e^{j\omega t}}^{f(\omega)} d\omega$$

$$= \frac{1}{2j} \left(\overbrace{e^{jt}}^{f(1)} - \overbrace{e^{-jt}}^{f(-1)} \right) + \frac{0.5}{2j} \left(\overbrace{e^{j20t}}^{f(20)} - \overbrace{e^{-j20t}}^{f(-20)} \right)$$

$$\therefore \hat{X}(j\omega) = j\pi \left(-\delta(\omega - 1) + \delta(\omega + 1) + 0.5(\delta(\omega - 20) - \delta(\omega + 20)) \right)$$

QUIZ I - FOURIER TRANSFORM

2. If

$$y(t) = \frac{dx}{dt} + \frac{d^2x(t)}{dt^2} + 3x(t),$$

and all initial conditions are zero, what is the Fourier transform of $y(t)$?

$$\hat{Y}(j\omega) = j\omega \hat{X}(j\omega) + (j\omega)^2 \hat{X}(j\omega) + 3\hat{X}(j\omega)$$

(A) $\hat{Y}(j\omega) = (j\omega + 1 + 3) \hat{X}(j\omega)$

(B) $\hat{Y}(j\omega) = ((j\omega) + (j\omega)^2 + 3) \hat{X}(j\omega)$

(C) $\hat{Y}(j\omega) = ((j\omega) + (j\omega)^2 + 3)$

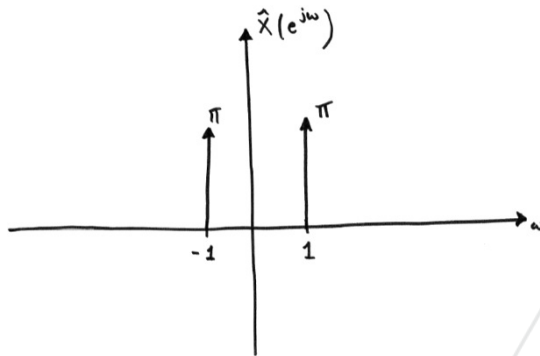
SOLUTION:

$$\begin{aligned} \mathcal{F}^{-1}\{y(t)\} &= \mathcal{F}^{-1}\{\dot{x} + \ddot{x} + 3x\} \\ &= \mathcal{F}^{-1}\{\dot{x}\} + \mathcal{F}^{-1}\{\ddot{x}\} + 3\mathcal{F}^{-1}\{x\} \quad \left. \begin{array}{l} \text{using linearity} \\ \text{using differentiation property} \\ \text{and zero initial conditions} \end{array} \right\} \\ &= j\omega \mathcal{F}^{-1}\{x\} + (j\omega)^2 \mathcal{F}^{-1}\{x\} + 3\mathcal{F}^{-1}\{x\} \\ &= (j\omega + (j\omega)^2 + 3) \mathcal{F}^{-1}\{x\} \end{aligned}$$

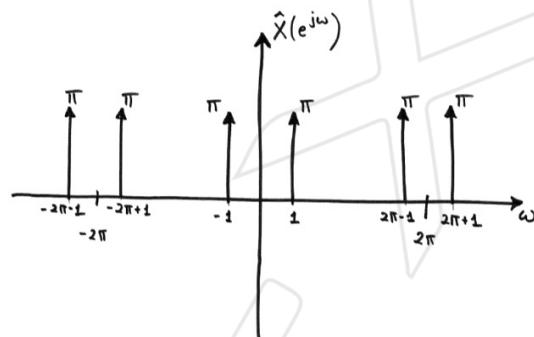
$$\hookrightarrow \hat{Y}(j\omega) = (j\omega + (j\omega)^2 + 3) \hat{X}(j\omega)$$

QUIZ I - FOURIER TRANSFORM

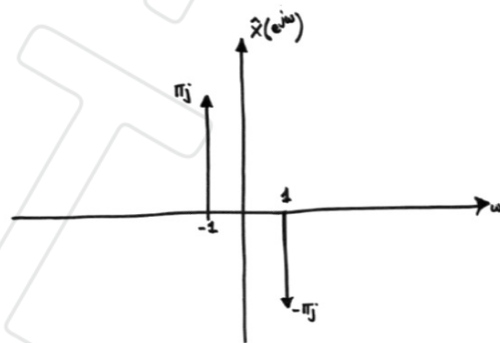
3. Which figure corresponds to the discrete-time Fourier transform of $x[n] = \cos[n]$?



(A)



(B)



(C)

QUIZ I - FOURIER TRANSFORM

4. What is $\hat{H}(e^{j\omega})$ for a system described by

$$y[n] + y[n-1] = x[n] - 2x[n-2]$$

(A) $\hat{H}(e^{j\omega}) = \frac{1 + e^{-j\omega}}{1 - 2e^{-2j\omega}}$

(B) $\hat{H}(e^{j\omega}) = \frac{1 - e^{-2j\omega}}{1 - e^{-j\omega}}$

(C) $\hat{H}(e^{j\omega}) = \frac{1 - 2e^{-2j\omega}}{1 + e^{-j\omega}}$

$$\hat{Y}(e^{j\omega})(1 + e^{-j\omega}) = \hat{X}(e^{j\omega})(1 - 2e^{-2j\omega})$$

$$\hat{H}(e^{j\omega}) = \frac{\hat{Y}(e^{j\omega})}{\hat{X}(e^{j\omega})}$$

$$\hat{Y}(e^{j\omega}) + e^{-j\omega} \hat{Y}(e^{j\omega}) = \hat{X}(e^{j\omega}) - 2e^{-2j\omega} \hat{X}(e^{j\omega})$$

$$(1 + e^{-j\omega}) \hat{Y}(e^{j\omega}) = (1 - 2e^{-2j\omega}) \hat{X}(e^{j\omega})$$

$$\hat{H}(e^{j\omega}) = \frac{\hat{Y}(e^{j\omega})}{\hat{X}(e^{j\omega})} = \frac{(1 - 2e^{-2j\omega})}{(1 + e^{-j\omega})}$$